Foundations of Full House Methodology

We now motivate the structure of the Full House Model. Let N_i denote the size of population i. We will suppose the every individual $k = 1, \ldots, N_i$ has an underlying talent value $X_{i,1}, \ldots, X_{i,N_i} \overset{\text{iid}}{\sim} F_X$. We denote $X_{i,(j)}$ as the jth ordered latent aptitude in year i. We define $g_i(\cdot, \cdot)$ as the system inclusion function, where $g_i(X_{i,j}, \mathbf{X}_{i,-j}) = 1$ indicates that individual j is included in the ith system, and $g_i(X_{i,j}, \mathbf{X}_{i,-j}) = 0$ indicates that individual j is not included in the ith system. Let $\mathbf{X}_{i,-j}$ be the vector of all latent aptitudes not including the component j.

We now suppose that an individual j included in system i has an observed statistics $Y_{i,j}$. We suppose that $Y_{i,1}, \ldots, Y_{i,n_i} \overset{iid}{\sim} F_{Y_i}$ where F_{Y_i} will be continuous, and $n_i < N_i$ is the number of individuals included in system i. We denote $Y_{i,(j)}$ as the jth ordered statistic for system i.

The key to the Full House Model is connecting talent values X with observed statistics Y. To do this we assume that we can form pairs $(Y_{i,(j)}, X_{i,(N_i - n_i + j)})$, $j = 1, \ldots, n_i$. Needs to be more general.

We now demonstrate how the Full House Model works in a parametric setting. Consider the pair $(Y_{i,(j)}, X_{i,(N_i - n_i + j)})$ (Needs to be more general, this currently corresponds to the case where all the highest latent aptitudes are in the system.) and suppose that the distribution corresponding to $Y_{i,j}$ from the ith system is known to belong to a continuous parametric family indexed by unknown parameter θ_i, and let $F_{Y_i}(\cdot \mid \theta_i)$ be a parametric CDF with parameter $\theta_i \in \mathbb{R}^{p_{Y_i}}$. We can estimate θ_i with $\hat{\theta}_i$ and plug the estimator into the CDF $F_Y(\cdot \mid \hat{\theta}_i)$.

In order to connect the $Y_{i,(j)}$ and $X_{i,(N_i - n_i + j)}$ and obtain an estimate of the underlying talents, we will make use of the following classical order statistics properties,

$$F_{Y_i}\left(Y_{i,(j)} \mid \theta_i\right) \sim U_{i,(j)}, \quad F_{Y_i}\left(Y_{i,(j)} \mid \hat{\theta}_i\right) \approx U_{i,(j)},$$
$$F_{Y_{i,(j)}}\left(Y_{i,(j)} \mid \theta_i\right) \sim U_{i,j}, \quad F_{Y_{i,(j)}}\left(Y_{i,(j)} \mid \hat{\theta}_i\right) \approx U_{i,j},$$

where $\approx$ means approximately distributed, $\sim$ means distributed as, $U_{i,j} \sim U(0,1)$, and $U_{i,(j)} \sim \text{Beta}(j, n_i + 1 - j)$ and the quality of the approximation in the right hand side depends upon the estimator $\hat{\theta}_i$ and the sample size.

We now connect the order statistics to the underlying talent distribution that comes from a population with $N_i \geq n_i$ observations when F_X is known. This connection is established with

$$F^{-1}_{X_{i,(N_i - n_i + j)}}\left(F_{U_{i,(j)}}\left(F_{Y_i}\left(Y_{i,(j)} \mid \theta_i\right)\right)\right) \sim F^{-1}_{X_{i,(N_i - n_i + j)}}\left(F_{U_{i,(j)}}\left(U_{i,(j)}\right)\right) = X_{i,(N_i - n_i + j)}. \quad (1)$$

We estimate the above with

$$F^{-1}_{X_{i,(N_i-n_i+j)}}\left(F_{U_{i,(j)}}\left(F_{Y_i}\left(Y_{i,(j)} \mid \hat{\theta}_i\right)\right)\right) \approx F^{-1}_{X_{i,(N_i-n_i+j)}}\left(F_{U_{i,(j)}}\left(U_{i,(j)}\right)\right) = \widehat{X}_{i,(N_i-n_i+j)}. \quad (2)$$

Similarly, we can obtain Y from X. First note that $F_X\left(X_{i,(N_i-n_i+j)}\right) \sim \text{Beta}(N_i-n_i+j, n_i-j+1)$ for $j = 1, \ldots, n_i$ where, for simplicity, we we denote $V(i,j) \stackrel{d}{=} \text{Beta}(N_i-n_i+j, n_i-j+1)$. Now,

$$F^{-1}_{Y_{i,(j)}}\left(F_{V(i,j)}\left(F_X\left(X_{i,(N_i-n_i+j)}\right)\right)|\theta_i\right) = Y_{i,(j)}, \quad (3)$$

where we estimate the above with $F^{-1}_{Y_{i,(j)}}\left(F_{V(i,j)}\left(F_X\left(\widehat{X}_{i,(N_i-n_i+j)}\right)\right)|\hat{\theta}_i\right) = \widehat{Y}_{i,(j)}$ for $j = 1, \ldots, n_i$.

Not sure where cross-contextual comparisons are needed. We can now reverse engineer the process above to obtain estimated statistics in another context. Consider the the pair $(Y_{i,(j)}, X_{i,(N_i-n_i+j)})$ in the ith population. We first put the aptitude value $X_{i,(N_i-n_i+1)}$ obtained by (2) in a new population m and reverse the process to obtain the hypothetical statistics in system m, which we will denote as $Y_{i,j,m}$. More formally, $Y_{i,j,m}$ are computed as follows when baseball statistics are estimated parametrically,

$$\widehat{Y}_{i,j,m} = F^{-1}_{Y_m}\left(F^{-1}_{U_{m,(l_{i,j,m})}}\left(F_{X_{m,(N_m-n_m+l_{i,j,m}))}}\left(\widehat{X}_{i,(N_i-n_i+j)}\right)\right)|\hat{\theta}_m\right), \quad (4)$$

where $l_{i,j,m}$ is the rank of $\widehat{X}_{i,(N_i-n_i+j)}$ among the values

$$\left\{\widehat{X}_{i,(N_i-n_i+j)}, \widehat{X}_{m,(N_m-n_m+t))} : t = 1, \ldots, n_m\right\}.$$

The estimates (4) correspond to

$$Y_{i,j,m} = F^{-1}_{Y_m}\left(F^{-1}_{U_{m,(l_{i,j,m})}}\left(F_{X_{m,(N_m-n_m+l_{i,j,m})}}\left(X_{i,(N_i-n_i+j)}\right)\right)\right), \quad (5)$$

where $l_{i,j,m}$ is the rank of $X_{i,(N_i-n_i+j)}$ among the values $\left\{X_{i,(N_i-n_i+j)}, X_{m,(N_m-n_m+t))} : t = 1, \ldots, n_m\right\}$, and $X_{i,(N_i-n_i+j)}$ is computed via (1).

Paradoxes

Consider two populations $i = 1, 2$ with sizes N_1 and N_2, and systems of sizes n_1 and n_2. We define a paradox (or a Gould paradox) when

$$X_{1,(N_1)} > X_{2,(N_2)} \quad \text{and} \quad Y_{1,(n_1)} < Y_{2,(n_2)}$$

both occur. A central interest is evaluating the probability of such a paradox,

$$\text{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right). \quad (6)$$

As a simple illustrative example of some mechanics concerning paradoxes, suppose that $Y_{1,j} < Y_{2,k}$ a.s. Then (6) is equivalent to $\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}\right)$, and we have $\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}\right) \to 1$ as $N_1 \to \infty$.

Consider the setting where $g_i\left(X_{i,j}, \mathbf{X}_{i,-j}\right) = 1\left(X_{i,j} \geq X_{i,(N_i - n_i + 1)}\right)$. Then (3) gives

$$F^{-1}_{Y_{i,(n_i)}}\left(F_{V(i,n_i)}\left(F_X\left(X_{i,(N_i)}\right)\right)|\theta_i\right) = Y_{i,(n_i)} \quad \text{for } i = 1, 2.$$

Let $f_i(\bullet) = F^{-1}_{Y_{i,(n_i)}}\left(F_{V(i,n_i)}\left(F_X\left(\bullet\right)\right)|\theta_i\right)$ and let $g_i(\bullet) = f_i^{-1}(\bullet)$. This construction yields,

$$\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, f_1\left(X_{1,(N_1)}\right) < f_2\left(X_{2,(N_2)}\right)\right).$$

Let $h_{i,k}(\bullet) = (g_i \circ f_k)(\bullet)$ when such an operation is defined. Then,

$$\begin{aligned}\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) &= \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, X_{1,(N_1)} < h_{1,2}\left(X_{2,(N_2)}\right)\right) \\ &= \mathrm{P}\left(X_{2,(N_2)} < X_{1,(N_1)} < h_{1,2}\left(X_{2,(N_2)}\right)\right).\end{aligned}$$